MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/348A, dated 9 May 1985, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE, delete and substitute as printed above.

* 1.2.3, add the following case outline:

"X C-2A (20-terminal, .350" x .350"), square chip carrier package

PAGE 2

Footnote 2/, delete in its entirety.

PAGE 3

TABLE I, low-level input, off-state low-output current, device type 02, max limits column: Delete "-1.6" and substitute "-1.0". Device type 03, max limits column: Delete "-1.6" and substitute "-1.2".

* TABLE I, low-level output voltage V_{OL}, device type column: Delete "ALL" and substitute "01, 02".

* TABLE I, low-level input, off-state low-output current, device type 01, min/max limits column: Delete "-.42/-1.0" and substitute "-.06/-1.6". Device type 02: Delete "-.03/-1.6" and substitute "-.06/-1.6".

* TABLE I, low-level input current I_{IL2}, device type 02: Delete in its entirety.

PAGE 5

* TABLE I, t_{PZH1}, max limits column, device type 03: Delete "9.5" and substitute "11.0".

* TABLE I, t_{PZH2}, max limits column, device type 03: Delete "9.5" and substitute "11.0".

PAGE 6

* TABLE II, additional electrical subgroups for group C periodic inspections, delete in its entirety.

* TABLE II, Class B devices column, group A test requirements, add "10,11"

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* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	Reprinted without change
12	12
12a	New

FSC 5962
AMSC N/A

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PAGE 7

4.2a.(1): Delete "D or E" and substitute "D, E, or F".

* 4.4.3b: Delete in its entirety.

4.4.3c.1: Delete "D or E" and substitute "D, E, or F".

PAGE 15

TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 16

TABLE III, test nos. 49 through 52, $\overline{EAB}$ column: Delete "0.5 V" and substitute "5.5 V".

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 17

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 18

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 19

* Footnote 1/, min/max limits column B, I_{IOZL}: Delete "-.03/-1.0" and substitute "-.06/-1.6".

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

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* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 21

TABLE III, test no. 48, $\overline{EAB}$ column: Add "0.5 V".

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 22

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 23

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

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PAGE 24

* Footnote 1/, delete in its entirety and substitute the following:

"1/ I_{IOZL}, I_{IL1} limits shall be as follows:

Test	Min/Max limits in (ma) for circuit		
	A	B	C
	Fairchild	Motorola	Signetics
I_{IL1}	-.42/-1.0	-.03/-1.0	-.03/-1.0
I_{IOZL}	-.67/-1.6	-.06/-1.6	-.06/-1.6

".

* TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

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* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 26

* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 27

TABLE III, symbol column: Delete "I_{IOZL}" and "I_{IOZH}" and substitute "I_{OZL}" and "I_{OZH}" respectively.

* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 28

* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 29

* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 30

TABLE III, test nos. 223 through 238, max limits column: Delete "7.5" and substitute "8.5"

* TABLE III, terminal designations, delete "Cases R, S, 2" and substitute "Cases R, S, 2, X".

PAGE 31

TABLE III, symbols t_{PZH1} and t_{PZH2}, max limits column: Delete "9.5" and substitute "11.0".

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PAGE 33

TABLE IV, device type 03, column C: Add "X".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

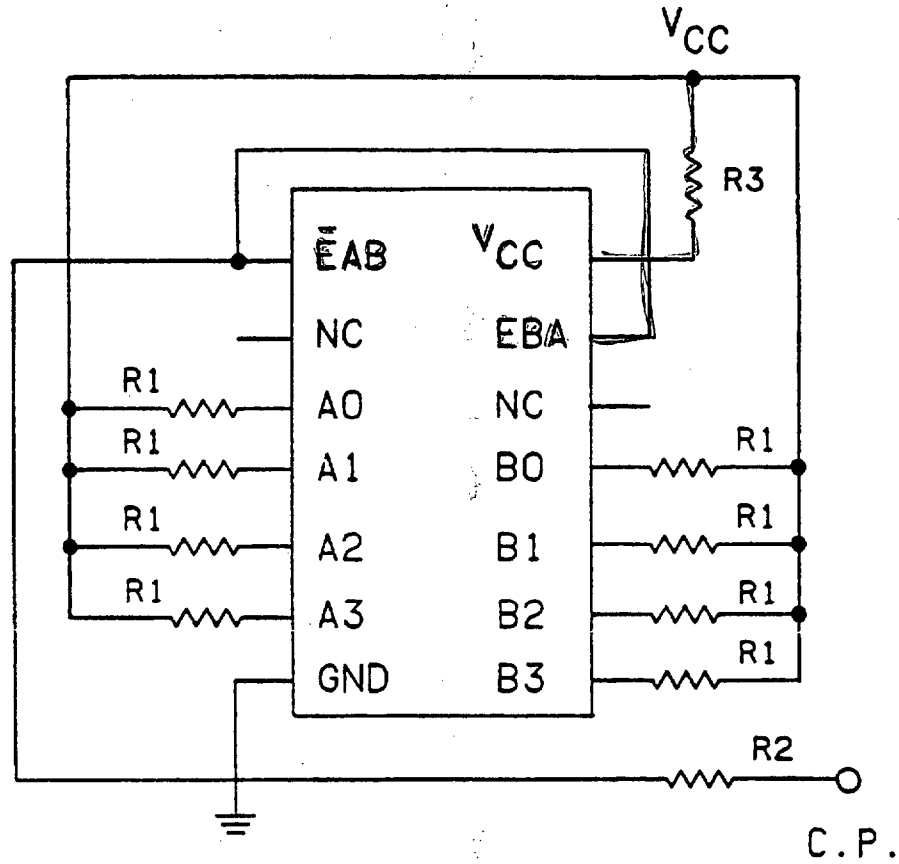
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1057)

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Device types 01 and 02



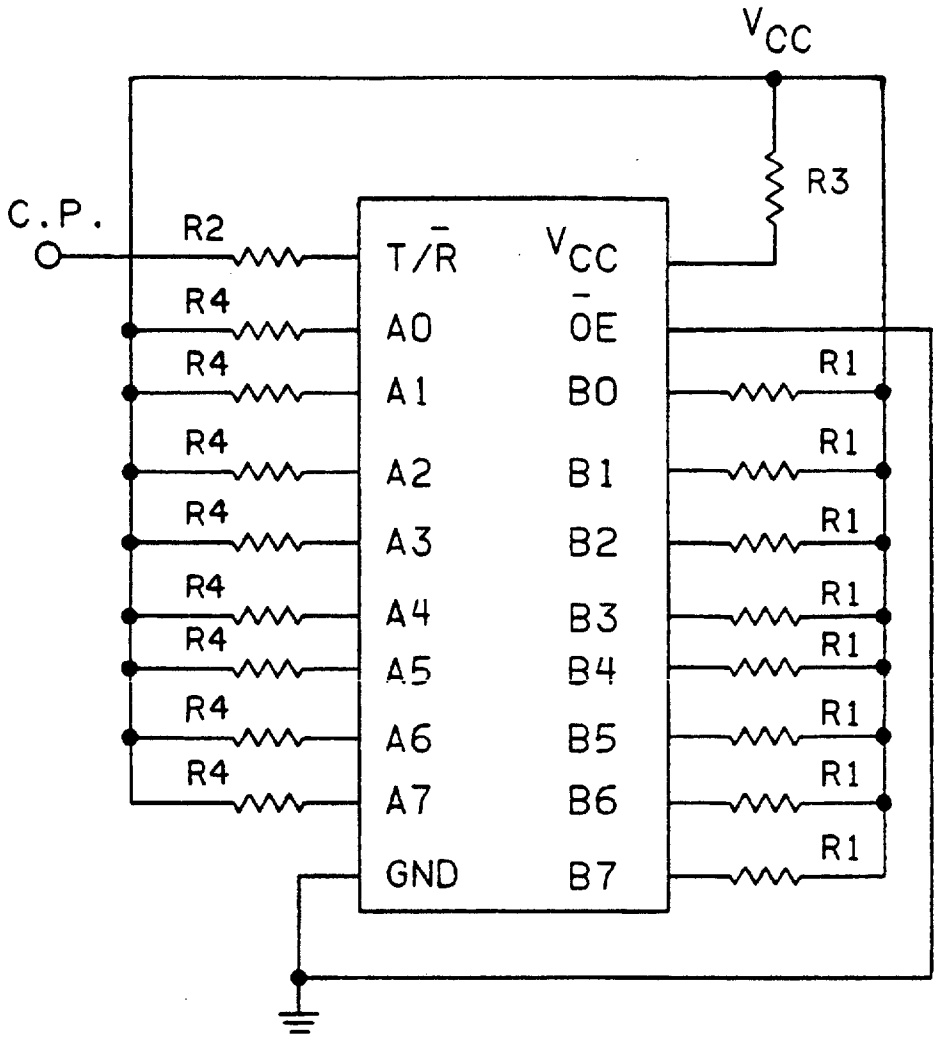
NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%.
   V_{IL} = -0.5 V minimum to 0.8 maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%, R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits.

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Device type 03

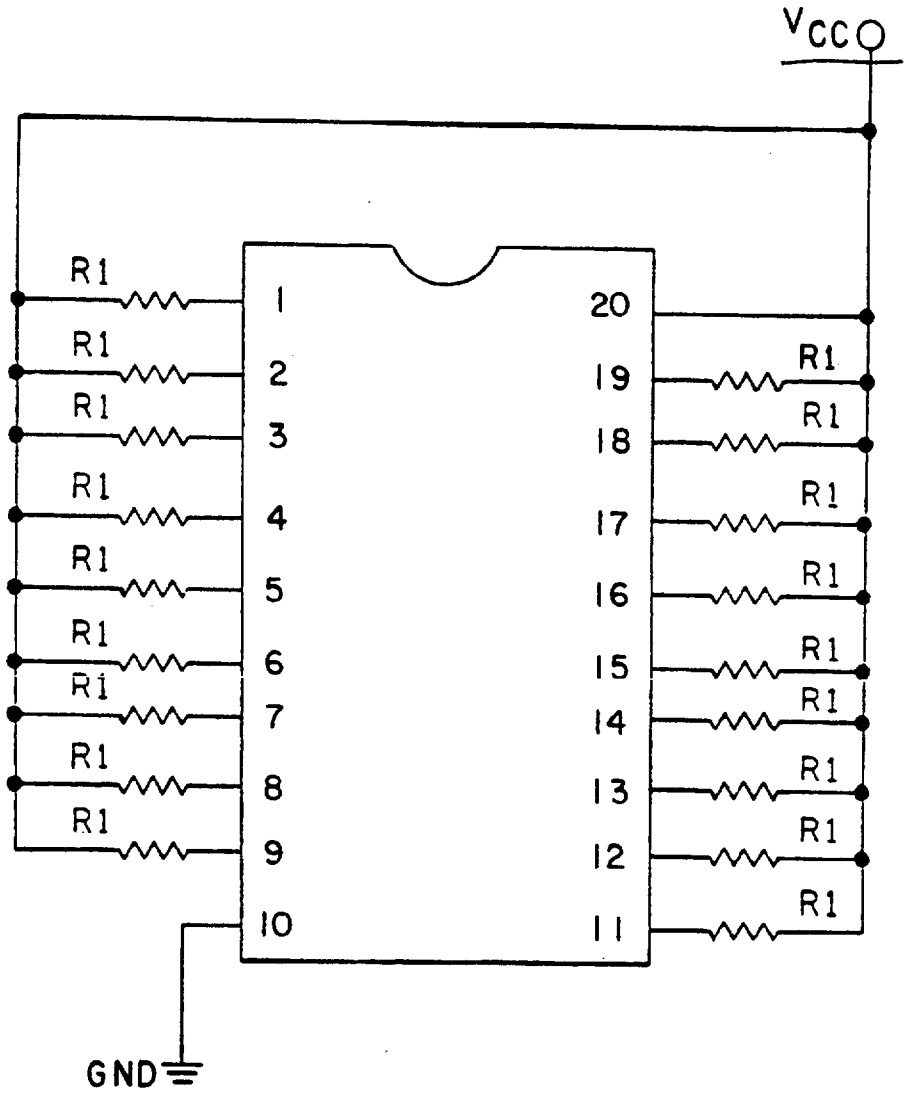


NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 100Ω ±5%, R2 = 51Ω ±5%, R4 = 200Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits - Continued.

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Device type 03



HIGH TEMPERATURE BURN-IN

NOTES:
1. R1 = 2,000 ohms.
2. V_{CC} shall be high enough to insure 5.0 V at the V_{CC} terminal.
3. High temperature (175°C) burn-in condition F (optional circuit) only.

FIGURE 3. Burn-in and life test circuits - Continued.

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